

Lab 5: Web API CRUD Operation (PUT)

Objective

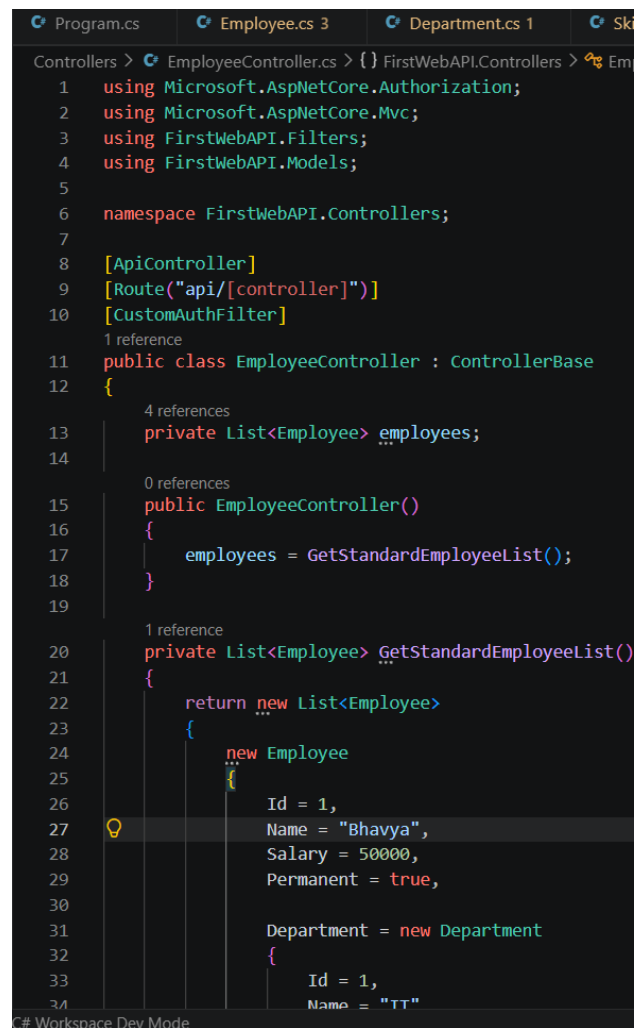
To implement the HTTP PUT action method in an ASP.NET Core Web API to update employee information using theFromBody attribute, validate employee IDs, return appropriate BadRequest responses for invalid IDs, and test the API using Swagger and Postman.

Commands Used

dotnet run

Code

Program.cs



```
Program.cs Employee.cs 3 Department.cs 1 Sk
Controllers > EmployeeController.cs > {} FirstWebAPI.Controllers > Em
1 using Microsoft.AspNetCore.Authorization;
2 using Microsoft.AspNetCore.Mvc;
3 using FirstWebAPI.Filters;
4 using FirstWebAPI.Models;
5
6 namespace FirstWebAPI.Controllers;
7
8 [ApiController]
9 [Route("api/[controller]")]
10 [CustomAuthFilter]
11 1 reference
12 public class EmployeeController : ControllerBase
13 {
14     4 references
15     private List<Employee> employees;
16
17     0 references
18     public EmployeeController()
19     {
20         employees = GetStandardEmployeeList();
21     }
22
23     1 reference
24     private List<Employee> GetStandardEmployeeList()
25     {
26         return new List<Employee>
27         {
28             new Employee
29             {
30                 Id = 1,
31                 Name = "Bhavya",
32                 Salary = 50000,
33                 Permanent = true,
34                 Department = new Department
35                 {
36                     Id = 1,
37                     Name = "TT"
38                 }
39             }
40         }
41     }
42 }
```

C# Workspace Dev Mode

Output

```
type.
C:\Users\laksh\FirstWebAPI\Filters\CustomAuthFilter.cs(18,18): warning CS8602: Dereference of possibly null value.
Build succeeded with 7 warning(s) in 1.6s
C:\Users\laksh\FirstWebAPI> dotnet run
Using launch settings from C:\Users\laksh\FirstWebAPI\Properties\launchSettings.json..
Building...
info: Microsoft.Hosting.Lifetime[14]
Now listening on: http://localhost:5074
info: Microsoft.Hosting.Lifetime[0]
Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
Content root path: C:\Users\laksh\FirstWebAPI
```

invalid request - no Auth token

Response headers

```
content-type: text/plain; charset=utf-8
date: Fri, 24 Jul 2026 14:10:24 GMT
server: Kestrel
transfer-encoding: chunked
```

Responses

Code	Description	Links
200	OK	No links
Media type: text/plain		
Controls: Accept header		
Example Value Schema		
<pre>{ "id": 0, "name": "string", "salary": 0, "permanent": true, "department": { "id": 0, "name": "string" }, "skills": [{ "id": 0, "name": "string" }], "dateOfBirth": "2026-07-24T14:10:25.360Z" }</pre>		
500	Internal Server Error	No links

Swagger UI interface showing a POST request to `/api/Employee`.

Parameters: No parameters.

Request body: application/json

Edit Value | Schema

```
{
  "id": 0,
  "name": "string",
  "salary": 0,
  "permanent": true,
  "department": {
    "id": 0,
    "name": "string"
  },
  "skills": [
    {
      "id": 0,
      "name": "string"
    }
  ],
  "dateOfBirth": "2026-07-24T14:11:00.785Z"
}
```

Execute **Clear**

Responses

Curl

Windows taskbar at the bottom shows the time as 19:41 on 24-07-2026.